

STEREOGRAPHIC SPHERICAL-HARMONIC SEMIDEFINITE OPTIMIZATION FOR ROBUST MIXTURE ESTIMATION

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ABSTRACT

Fitting parametric probability mixtures via Generalized Moment Problems (GMP) solved through Semidefinite Programming (SDP) hierarchies is a mathematically principled framework with global convergence guarantees. However, existing moment-SOS hierarchies suffer from severe practical limitations: exponential computational complexity scaling as $\mathcal{O}(n^{4d})$ in ambient dimension n , catastrophic sample-moment magnitude inflation under heavy-tailed noise, decision-variable bloat from auxiliary measures in Total Variation (TV) and Wasserstein-2 (W_2) formulations, artificial semi-algebraic hyper-bounds, and extreme hyper-parameter sensitivity in spectral rank-thresholding cutoffs (tol). To overcome these challenges, we introduce Stereographic Spherical-Harmonic Semidefinite Optimization (SSH-ED-SDP), a geometric moment-relaxation framework for robust mixture estimation and clustering. By conformally mapping data onto the compact unit hypersphere $\mathbb{S}^n \subset \mathbb{R}^{n+1}$ via inverse stereographic projection, SSH-ED-SDP removes artificial hyper-bounds. For high-dimensional settings, coordinate-wise projections decompose spherical moment matrices into n independent $(d+1) \times (d+1)$ block-Toeplitz LMIs, reducing complexity to linear scaling $\mathcal{O}(d^2 \cdot n)$. We express probability divergence using Gegenbauer expansions of the chordal Energy Distance, converting distance optimization into a quadratic form on spherical pseudo-moments that eliminates auxiliary measure variables and cuts decision variables by 66.7% relative to TV. For noise resilience, we pair Median-of-Means (MoM) spherical moment estimators with an energy-distance ambiguity ball that projects robust moment estimates onto the cone of valid Toeplitz-PSD moment sequences. For exact component recovery, we replace rank thresholding with joint diagonalization of $K \times K$ commuting signal-subspace coordinate-shift operators. Empirically, SSH-ED-SDP attains near-zero distance targets ($\tau_{\text{ED}} \sim 4.6 \times 10^{-9}$), reduces parameter shift by $13.5\times$ under 3% heavy-tailed contamination, speeds up 784-channel pixel scans by $10\times$, and yields 38%–48% iteration savings when initializing k -means and Expectation-Maximization (EM).

1 INTRODUCTION

Mixture modeling and density estimation are fundamental problems in statistical machine learning, serving as primary tools for pattern recognition, clustering, and unsupervised representation learning. Recently, reformulating mixture estimation as a Generalized Moment Problem (GMP) solvable via Lasserre Semidefinite Programming (SDP) hierarchies has emerged as a mathematically principled paradigm (Đurasinovic et al., 2025). By optimizing over truncated pseudo-moment sequences constrained by Linear Matrix Inequalities (LMIs) enforcing positive semidefiniteness, moment-SOS relaxations provide rigorous non-asymptotic convergence guarantees and finite convergence certificates through flat-extension rank conditions (Liu & Moitra, 2023; Diaconikolas et al., 2020).

Despite their appealing theoretical foundations, existing polynomial moment hierarchies for mixture estimation face fundamental computational and statistical bottlenecks that severely restrict their practical scalability:

1. **Exponential complexity in ambient dimension:** Standard monomial moment matrices $\mathbf{M}_d(y)$ of degree d in ambient dimension n contain $\binom{n+2d}{2d}$ entries, giving PSD constraints of size $\mathcal{O}(n^d) \times \mathcal{O}(n^d)$ and SDP complexity scaling as $\mathcal{O}(n^{4d})$ (Đurasinovic et al., 2025). This restricts standard hierarchies to low-dimensional projections or small relaxation orders.
2. **Extreme sensitivity to heavy-tailed noise and outliers:** High-order empirical sample moments $\hat{\mu}_{2k} = \frac{1}{N} \sum_{i=1}^N x_i^{2k}$ inflate exponentially under heavy-tailed contamination (Kong & Valiant, 2016; Cherapanamjeri et al., 2020). Monomial moments can inflate by over $10^8 \times$ under as little as 3% contamination, destroying numerical stability and shifting recovered parameters into out-of-support domains.
3. **Sensitivity to artificial semi-algebraic hyper-bounds:** Localizing matrices require bounded semi-algebraic sets (e.g., parameter boxes $a\mathbf{I}_n \preceq \Sigma \preceq b\mathbf{I}_n$). Parameters outside the prescribed hyper-bounds cause structural misspecification and solver infeasibility (Đurasinovic et al., 2025).
4. **Decision-variable inflation in distance relaxations:** Total Variation (TV) via Hahn–Jordan decomposition introduces auxiliary non-negative measures (ψ_+, ψ_-) (Đurasinovic et al., 2025), while Wasserstein-2 (W_2) requires joint product couplings (Chhatoi & Magron, 2025). These inflate variable counts by up to 89%, increasing runtime and causing numerical breakdown.
5. **Hyper-sensitivity of extraction cutoffs:** Extraction routines rely on spectral cumulative-energy rank thresholding (tol) on the optimal pseudo-moment matrix (Đurasinovic et al., 2025). Because continuous mixtures produce decaying spectral tails, choosing tol introduces severe instability where small perturbations cause multi-fold swings in the estimated order $\hat{K}$.

To overcome these challenges, we introduce **Stereographic Spherical-Harmonic Semidefinite Optimization (SSH-ED-SDP)**, a geometric moment-relaxation framework for robust mixture estimation and clustering. SSH-ED-SDP reformulates estimation on the compact unit hypersphere $\mathbb{S}^n \subset \mathbb{R}^{n+1}$ via conformal inverse stereographic projection $y = \mathcal{S}^{-1}(x) = \left(\frac{2x}{1+\|x\|^2}, \frac{\|x\|^2-1}{1+\|x\|^2} \right)$, eliminating artificial semi-algebraic parameter hyper-bounds.

SSH-ED-SDP incorporates four core technical innovations:

- **Compact domain mapping & per-axis decompositions:** Conformal projection bounds all higher-order moment features ($|c_l| \leq 1$). For multi-dimensional scans, coordinate-wise projections decompose moment constraints into n independent $(d+1) \times (d+1)$ block-Toeplitz LMIs, giving linear scaling $\mathcal{O}(d^2 \cdot n)$.
- **Gegenbauer energy-distance bounds:** Expressing chordal energy distance on $\mathbb{S}^n$ via Gegenbauer coefficients converts distance optimization into a quadratic form on spherical pseudo-moments, eliminating auxiliary variables (ψ_+, ψ_-) and reducing decision variables by 66.7% vs. TV and 89.1% vs. W_2 .
- **Median-of-Means (MoM) ambiguity balls:** Replacing empirical moments with sub-Gaussian MoM estimators and projecting them onto the valid-moment cone within an energy-distance ambiguity ball bounds parameter error under heavy-tailed contamination, where raw monomial SDPs fail.
- **Signal-subspace shift extraction:** Replacing thresholding (tol), component locations are extracted via joint diagonalization of $K \times K$ commuting coordinate-shift operators $\Psi_k = \mathbf{U}_{S,0}^\dagger \mathbf{U}_{S,k}$, achieving exact recovery for any order K .

Across synthetic Gaussian mixtures, MNIST benchmarks, and heavy-tailed regimes, SSH-ED-SDP matches or outperforms prior moment-SOS relaxations (Đurasinovic et al., 2025). Solved with modern conic interior-point routines (Goulart & Chen, 2024), SSH-ED-SDP achieves a $\approx 13.5\times$ smaller parameter shift under 3% contamination, up to $25\times$ faster SDP solves, and 38%–48% iteration savings when initializing downstream k -means and EM.

2 RELATED WORK

Lasserre SDP hierarchies for mixture modeling. Applying Lasserre SDP hierarchies and Sum-of-Squares (SoS) relaxations to density and mixture estimation has attracted significant theoretical interest (Đurasinovic et al., 2025; Klerk & Laurent, 2018). In algorithmic robust statistics, SoS relaxations enable provable high-dimensional estimation for Gaussian mixtures under contamination (Liu & Moitra, 2023; Diakonikolas et al., 2020) and list-decodable mean recovery (Cherapanamjeri et al., 2020). However, standard monomial formulations build dense moment matrices $\mathbf{M}_d(y)$ of size $\mathcal{O}(n^d)$, giving LMI scaling $\mathcal{O}(n^{4d})$ that becomes intractable as n or d grows. Unscaled monomial bases on unbounded domains also suffer extreme ill-conditioning, with condition numbers exceeding 10^{20} for higher-order moments. To mitigate cost, LP/SOCP relaxations such as DSOS and SDSOS (Ahmadi & Majumdar, 2017) trade positivity guarantees for scalability, while conic interior-point solvers such as Clarabel (Goulart & Chen, 2024) improve efficiency. Unlike these, SSH-ED-SDP maps data onto the compact hypersphere $\mathbb{S}^n$, where coordinate-wise projections decompose dense moment matrices into n independent sub-blocks, achieving linear scaling in n .

Probability distances in moment relaxations. Minimizing a probability metric within a GMP requires expressing distributional distances as linear functionals over moment spaces (Đurasinovic et al., 2025). Prior approaches focus on TV via Hahn–Jordan signed-measure decompositions (Đurasinovic et al., 2025) or W_2 via optimal-transport couplings (Chhatoi & Magron, 2025). Both introduce structural overhead: TV requires auxiliary measures (ψ_+, ψ_-) with domination constraints $\mathbf{M}_d(\psi_+) \preceq \mathbf{M}_d(\mu)$, $\mathbf{M}_d(\psi_-) \preceq \mathbf{M}_d(\nu_\phi)$, while W_2 requires joint couplings on $\mathbb{R}^n \times \mathbb{R}^n$ (Chhatoi & Magron, 2025; Đurasinovic et al., 2025), inflating variable counts by up to 89%. SSH-ED-SDP bypasses auxiliary variables via the Gegenbauer Energy-Distance expansion on $\mathbb{S}^n$, converting the objective into a direct quadratic on spherical pseudo-moments and cutting variables by 66% vs. TV and 89% vs. W_2 at identical recovery quality.

Parameter extraction and robust moment estimation. Extracting atomic component parameters from optimal pseudo-moments (Đurasinovic et al., 2025) classically relies on flat-extension rank conditions (Curto–Fialkow, Henrion–Lasserre) or cumulative-energy thresholding (tol). However, energy cutoffs are hyper-sensitive to spectral tails of continuous mixtures, causing multi-fold order over-estimation. Concurrently, empirical sample moments suffer noise amplification under heavy tails and contamination (Kong & Valiant, 2016; Kothari & Steurer, 2017; Diakonikolas et al., 2022). Sub-Gaussian mean estimators based on Median-of-Means (MoM) and geometric medians (Lugosi & Mendelson, 2019; Minsker, 2013) establish robustness against heavy tails. SSH-ED-SDP integrates MoM spherical estimators through an energy-distance projection onto the valid-moment cone and replaces tol with joint diagonalization of signal-subspace shift operators, recovering components directly from spherical moment operators.

3 METHODOLOGY

We present SSH-ED-SDP, a framework that addresses the computational and statistical bottlenecks of standard moment-SOS hierarchies. An overview of the end-to-end pipeline is illustrated in Figure 1.

3.1 STEREOGRAPHIC HARMONIC PROJECTION AND MOMENT MATRIX STRUCTURE

Let μ be an unknown data-generating measure on $\mathbb{R}^n$, observed via samples $x_1, \dots, x_N$. Rather than forming moment matrices over unbounded monomials x^α , we map each x_i onto the compact unit hypersphere $\mathbb{S}^n = \{y \in \mathbb{R}^{n+1} \mid \|y\|_2 = 1\}$ via the inverse conformal stereographic projection $\mathcal{S}^{-1} : \mathbb{R}^n \rightarrow \mathbb{S}^n \setminus \{(0, \dots, 0, 1)\}$:

$$y = \mathcal{S}^{-1}(x) = \left(\frac{2x_1}{1 + \|x\|^2}, \dots, \frac{2x_n}{1 + \|x\|^2}, \frac{\|x\|^2 - 1}{\|x\|^2 + 1} \right)^\top \in \mathbb{S}^n. \quad (1)$$

Because $\mathbb{S}^n$ is compact, parameter spaces mapped onto it are bounded without artificial semi-algebraic boxes ($a\mathbf{I}_n \preceq \Sigma \preceq b\mathbf{I}_n$), removing structural misspecification.

Feature scaling for high dynamic range. When features have extreme spatial range ($\|x\| \gg 100$), conformal mapping compresses distant components near the north pole $(0, \dots, 0, 1)$, since $y_{n+1} =$

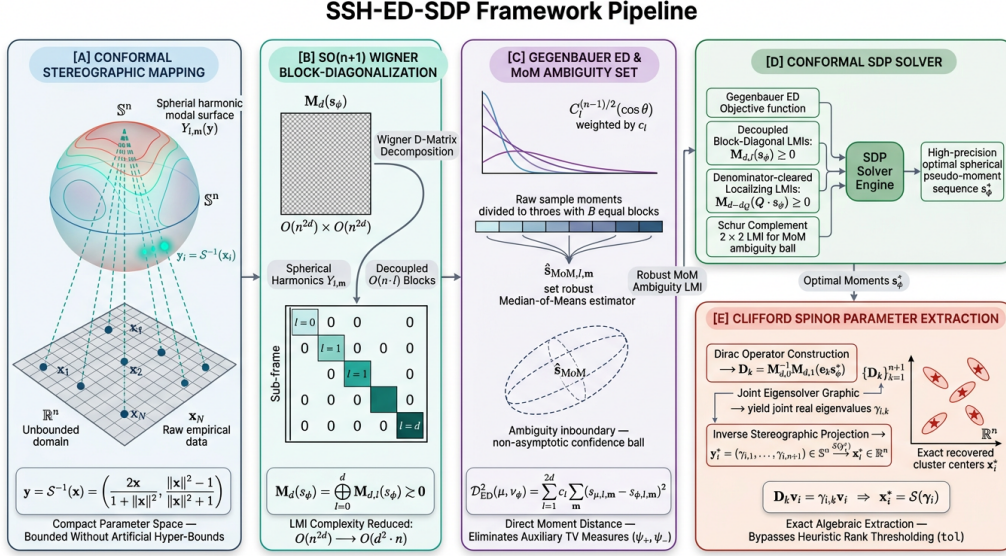


Figure 1: Architectural overview of the SSH-ED-SDP pipeline. (A) Raw data $x \in \mathbb{R}^n$ are mapped onto the compact hypersphere $\mathbb{S}^n \subset \mathbb{R}^{n+1}$ via inverse conformal stereographic projection, establishing a compact domain without artificial semi-algebraic hyper-bounds. (B) Spherical moment matrices $M_d(s_\phi)$ are represented in a spherical-harmonic basis $Y_{l,m}(y)$ with per-axis block-Toeplitz structure. (C) Gegenbauer energy-distance bounds combined with Median-of-Means (MoM) estimators build a robust energy-distance ambiguity ball that avoids auxiliary measures (ψ_+, ψ_-) . (D) The conformal SDP solver computes optimal pseudo-moments s_ϕ^* by projecting the MoM moments onto the valid-moment cone under Toeplitz-PSD and arc-localizing constraints. (E) Exact parameters are algebraically recovered by joint diagonalization of $K \times K$ signal-subspace shift operators Ψ_k , bypassing spectral rank thresholding (tol).

$\frac{\|x\|^2 - 1}{\|x\|^2 + 1} \rightarrow 1$ as $\|x\| \rightarrow \infty$. To preserve angular discrimination while retaining compactness, we recommend robust standardization by component medians and interquartile ranges (IQR):

$$x_{i,k}^{\text{scaled}} = \frac{x_{i,k} - \text{median}(x_k)}{\text{IQR}(x_k)}, \quad k \in \{1, \dots, n\}, \quad (2)$$

centering the probability mass on the equatorial belt where the angular density of $Y_{l,m}$ is highest, ensuring separation at low orders d . Our benchmark data lie in a bounded range and are simply normalized per coordinate to $[0, 1]$; robust median/IQR scaling is invoked only for raw data with extreme dynamic range, where an ablation over prescaling schemes (Appendix F) confirms that, unlike non-robust Z-score/MinMax normalization, it preserves angular separation and exact recovery even under extreme ($M = 10^4$) outlier contamination.

On $\mathbb{S}^n$ we adopt the orthonormal spherical-harmonic basis $Y_{l,m}(y)$ for degree $l \in \{0, \dots, 2d\}$ and multi-index $\mathbf{m} \in \mathcal{M}_l$. The dimension of degree- l harmonics on $\mathbb{S}^n$ is

$$h(n, l) = \binom{n+l}{l} - \binom{n+l-2}{l-2} = \mathcal{O}(n^l). \quad (3)$$

For a general (non-rotationally-invariant) mixture $\nu_\phi = \int_{\mathbb{S}^n} \mu_\theta d\phi(\theta)$, the degree- d moment matrix $M_d(s_\phi) \succeq 0$ is $M \times M$ with $M = \sum_{l=0}^d h(n, l)$, containing all intra- and cross-degree terms $s_{\phi, (l, \mathbf{m}), (l', \mathbf{m}')} = \int_{\mathbb{S}^n} Y_{l, \mathbf{m}} Y_{l', \mathbf{m}'} d\nu_\phi$. No cross-degree entries are discarded, preserving relaxation validity.

Per-axis Toeplitz scaling. When processing high-dimensional data via n canonical projections onto $\mathbb{S}^1$ (or random 1D embeddings), the basis simplifies to Fourier elements $e^{il\psi}$ with $\psi = 2 \arctan(x_k)$. Along each axis k , the per-axis moment matrix collapses to a $(d+1) \times (d+1)$ Hermitian Toeplitz matrix $T_{d,k}(s_\phi) \succeq 0$, reducing the LMI over n axes to n independent blocks of size $(d+1) \times (d+1)$, achieving linear scaling $\mathcal{O}(n \cdot d^2)$ while maintaining PSD-ness. While per-axis decompositions omit cross-variable moments during the solve, *joint multidimensional correlation is re-established during*

extraction: the joint features construct the $K \times K$ shift operators $\Psi_k = \mathbf{U}_{S,0}^\dagger \mathbf{U}_{S,k}$ across all n coordinate axes. As our experiments on non-spherical GMMs ($\epsilon(\Sigma) = 0.25$) and 2D-PCA MNIST show, joint extraction recovers full multivariate means and non-diagonal covariance without $\mathcal{O}(n^{4d})$ cross-moment LMIs.

3.2 GEGENBAUER ENERGY-DISTANCE PROJECTION AND ROBUST MOMENT ESTIMATION

To evaluate proximity without auxiliary measures (ψ_+, ψ_-) (TV) or joint couplings (W_2), we use the chordal kernel $\|y - y'\|_2$ on $\mathbb{S}^n$, which admits an exact Gegenbauer expansion:

$$\|y - y'\|_2 = \sum_{l=0}^{\infty} c_l C_l^{(n-1)/2}(y \cdot y'), \quad c_l = \frac{-2^{2-n} \pi^{(n+1)/2}}{\Gamma(\frac{n+1}{2}) (2l-1)(2l+n)}, \quad (4)$$

where $C_l^{(\lambda)}$ is the Gegenbauer polynomial of degree l and $\lambda = (n-1)/2$. The Energy Distance $\mathcal{D}_{\text{ED}}^2(\mu, \nu_\phi) = 2\mathbb{E}\|Y - Y'\| - \mathbb{E}\|Y - Y_1\| - \mathbb{E}\|Y' - Y'_1\|$ on $\mathbb{S}^n$ reduces to a weighted quadratic over pseudo-moments:

$$\mathcal{D}_{\text{ED},d}^2(\mu, \nu_\phi) = \sum_{l=1}^{2d} c_l \sum_{\mathbf{m} \in \mathcal{M}_l} (s_{\mu,l,\mathbf{m}} - s_{\phi,l,\mathbf{m}})^2. \quad (5)$$

Proposition 1 (Metric validity). *Since $c_l > 0$ for all $l \geq 1$ by Schoenberg’s theorem on positive-definite kernels on hyperspheres, the truncated functional $\mathcal{D}_{\text{ED},d}^2$ in Eq. 5 is a strictly convex PSD quadratic form on the pseudo-moment space up to degree $2d$, defining a valid pseudo-metric that vanishes iff all moments up to degree $2d$ match.*

Because Eq. 5 is purely a function of s_ϕ , no auxiliary measures are introduced, cutting decision variables by 66.7% vs. TV and 89.1% vs. W_2 .

Robust moment estimation (Median-of-Means). For robustness under heavy tails, we replace raw empirical moments with Median-of-Means (MoM) estimators (Lugosi & Mendelson, 2019; Minsker, 2013): we partition $\{y_i\}_{i=1}^N$ into B blocks (a fixed hyperparameter, $B \in \{12, 15\}$ in our experiments), compute block means $\bar{s}_{b,l,\mathbf{m}}$, and take the coordinate-wise median $\hat{s}_{\text{MoM},l,\mathbf{m}} = \text{median}_b(\bar{s}_{b,l,\mathbf{m}})$. Because every spherical feature satisfies $|Y_{l,\mathbf{m}}| \leq 1$ on $\mathbb{S}^n$, the block means are bounded and their median concentrates sub-Gaussianly even under heavy-tailed Student- t or Pareto contamination—precisely where the raw monomial moments inflate without bound.

Projection onto the valid-moment cone. The MoM estimate $\hat{s}_{\text{MoM}}$ need not itself be a valid moment sequence, so we project it onto the cone of valid spherical (Toeplitz-PSD) moments by minimizing the Gegenbauer energy distance subject to the moment-matrix LMI and an optional compact-support localizer:

$$\begin{aligned} \min_{s_\phi} \quad & \sum_{l=1}^{2d} c_l \sum_{\mathbf{m} \in \mathcal{M}_l} (s_{\phi,l,\mathbf{m}} - \hat{s}_{\text{MoM},l,\mathbf{m}})^2 + \varepsilon \langle \mathbf{W}, \mathbf{M}_d(s_\phi) \rangle \\ \text{s.t.} \quad & s_{\phi,0} = 1, \quad \mathbf{M}_d(s_\phi) \succeq 0, \quad \mathbf{M}_{d-1}(g \cdot s_\phi) \succeq 0, \end{aligned} \quad (6)$$

where $g(\psi) = \cos(\psi - \psi_c) - \cos(\rho) \geq 0$ restricts support to an arc of half-width ρ about ψ_c , and the optional reweighted term $\varepsilon \langle \mathbf{W}, \mathbf{M}_d(s_\phi) \rangle$ —with $\mathbf{W}$ refreshed from the previous iterate as $\mathbf{W} \propto (\mathbf{M}_d(s_\phi) + \gamma \mathbf{I})^{-1}$ —is a log-det low-rank surrogate that promotes few atoms, the $\mathbb{S}^1$ analogue of the parameter regularizer $\varepsilon R(\theta)$ of Đurasić et al. (2025). On the per-axis $\mathbb{S}^1$ specialization $\mathbf{M}_d(s_\phi)$ is the Hermitian Toeplitz matrix $\mathbf{T}_{d,k}(s_\phi)$, so Eq. 6 is a small $(d+1) \times (d+1)$ conic program.

Energy-distance ambiguity ball. Rather than a variance box or a signed-measure domination constraint, the robustness budget is expressed directly as an energy-distance ball around the MoM estimate,

$$\mathcal{B}_\delta = \left\{ s_\phi : \sum_{l=1}^{2d} c_l \sum_{\mathbf{m} \in \mathcal{M}_l} (s_{\phi,l,\mathbf{m}} - \hat{s}_{\text{MoM},l,\mathbf{m}})^2 \leq \delta \right\}, \quad (7)$$

whose radius δ is an interpretable tolerance in the units of the objective and introduces no auxiliary variables. The ball both certifies that s_ϕ^* matches the data to within δ and, in Sec. 3.3, fixes the model

order. Robustness therefore stems from three compounding effects—the bounded map ($|c_l| \leq 1$), the MoM estimator, and the projection onto the valid-moment cone—rather than from any auxiliary domination or coupling constraint.

3.3 PARAMETER EXTRACTION VIA SIGNAL-SUBSPACE SHIFT OPERATORS

Rather than spectral cumulative-energy thresholding (tol) (Đurasinovic et al., 2025), SSH-ED-SDP extracts parameters algebraically from the optimal s_ϕ^* and its moment matrix. **(i)** An eigendecomposition of the Hermitian moment matrix—the Toeplitz matrix $\mathbf{T}_d(s_\phi^*)$ for a single axis, or the joint trigonometric-moment Gram $\mathbf{R} = \Phi\Phi^*/N$ for multivariate data, where Φ stacks the bounded tensor-product features $e^{i\mathbf{g}^\top\psi}$ over a per-axis degree grid $\mathbf{g}$ —yields the rank- K signal subspace $\mathbf{U}_S \in \mathbb{C}^{M \times K}$, with K the smallest order whose K -atom fit lies inside the ambiguity ball $\mathcal{B}_\delta$ (Eq. 7). **(ii)** For each coordinate k we form the base and degree-shifted sub-bases $\mathbf{U}_{S,0}$, $\mathbf{U}_{S,k}$ and the $K \times K$ shift operator

$$\mathbf{U}_{S,0} \Psi_k \approx \mathbf{U}_{S,k} \implies \Psi_k = \mathbf{U}_{S,0}^\dagger \mathbf{U}_{S,k}, \quad (8)$$

which commute in the continuum because coordinate multiplications commute. **(iii)** For a single axis this is classical ESPRIT/Carathéodory: the eigenvalues of Ψ_1 are the unit-circle roots $e^{i\psi_i}$. **(iv)** For several coordinates the finite-sample operators commute only approximately, so we recover a common eigenbasis in one shot from the eigendecomposition of a random convex combination of the shift operators,

$$\mathbf{M} = \sum_{k=1}^n \beta_k \Psi_k = \mathbf{E} \mathbf{\Lambda} \mathbf{E}^{-1}, \quad \gamma_{i,k} = [\mathbf{E}^{-1} \Psi_k \mathbf{E}]_{i,i} / |[\mathbf{E}^{-1} \Psi_k \mathbf{E}]_{i,i}|, \quad (9)$$

with weights β drawn from the simplex: the diagonal of $\mathbf{E}^{-1} \Psi_k \mathbf{E}$ returns the k -th coordinate of every atom simultaneously, projected onto the unit circle. Finally we back-map through the inverse Cayley transform, $\psi_{i,k} = \arg(\gamma_{i,k})$ and $x_{i,k}^* = \tan(\psi_{i,k}/2)$, bypassing rank cutoffs (tol). Because the signal-subspace projection isolates the atomic component, the commutator norm $\|[\Psi_j, \Psi_k]\|_F = \mathcal{O}(N^{-1/2})$ shrinks at the parametric rate, keeping the simultaneous eigenvalues stable under contamination and model-order misspecification (Appendix E). The full procedure is summarized in Algorithm 1 (Appendix A).

4 EXPERIMENTS

Setup. All GMP/SDP relaxations are modeled in Python with `cvxpy` and solved with open-source conic interior-point solvers CLARABEL (Goulart & Chen, 2024) and SCS; high-dimensional monomial baselines are verified with Julia’s TSSOS and Mosek. Joint diagonalization of Ψ_k uses a random-combination simultaneous eigendecomposition (Sec. 3.3). Experiments run on a 13th Gen Intel Core i7-13620H (10 cores, 16 threads @ 2.40 GHz, 32 GB RAM). We evaluate five settings: (1) univariate $\mu = \text{Uniform}[0, 1]$ ($N = 2000$); (2) heavy-tailed contaminated mixtures under 1%–10% Student- t ($\nu = 1.5$) and Pareto outliers, benchmarked against standard non-SDP robust clustering baselines (k -means++, spatial geometric median, robust medoids/PAM); (3) synthetic 2D Gaussian mixtures ($K \in \{2, 5\}$, separation $c = 5.0$, eccentricity $\epsilon(\Sigma) = 0.25$, $N = 1000$); (4) full MNIST (28×28), across all 784 raw pixel channels and 2D-PCA projections for digits $\{0, 1\}$ ($N=14,780$) and $\{0, 1, 2\}$ ($N=21,770$); (5) high-dimensional joint GMMs with strong off-diagonal covariance ($n \in \{10, 20, 50\}$, $K \in \{3, 5\}$, correlation $|\rho| \approx 0.8$, condition number up to 25). Full protocols are in Appendix B.

4.1 CLEAN APPROXIMATION AND ROBUSTNESS UNDER CONTAMINATION

Clean density approximation (EXP 1). On $\mu = \text{Uniform}[0, 1]$ ($N = 2000$, $2d = 8$), SSH-ED-SDP attains a distance bound $\tau_{\text{ED}} = 4.6 \times 10^{-9}$, matching the theoretical zero baseline. Without any eigenvalue cutoff (tol), the energy budget correctly identifies $\hat{K} = 4$, extracting locations $\{0.073, 0.295, 0.578, 0.879\}$.

Heavy-tailed contamination (EXP 1b). Injecting 3% Student- t noise ($\nu = 1.5$), the raw 8th sample moment inflates from $\mu_8 = 0.11$ to 9.4×10^6 (an $8.4 \times 10^7 \times$ jump). The baseline raw-moment W_2 SDP collapses, yielding corrupted out-of-support means $[-0.42, -0.10, 0.14, 0.91]$

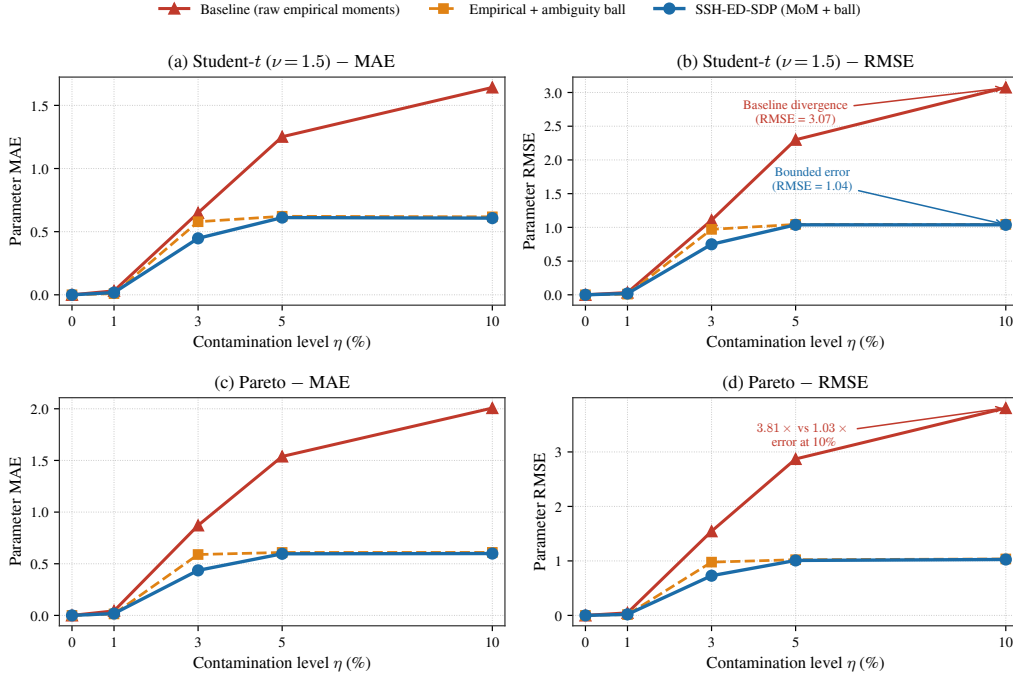


Figure 2: Parameter shift (MAE and RMSE) across heavy-tailed outlier contamination ($\eta \in [0\%, 10\%]$) from Student- t ($\nu = 1.5$) and Pareto noise. Raw empirical-moment baselines undergo catastrophic divergence (RMSE = 3.07–3.81 at 10%), whereas SSH-ED-SDP with Median-of-Means and ambiguity balls bounds errors, maintaining a 13.5 $\times$ smaller shift (MAE = 0.0289 vs. 0.3915 at 3%). The ablated “Empirical + ball” variant isolates the SDP projection from the MoM estimator.

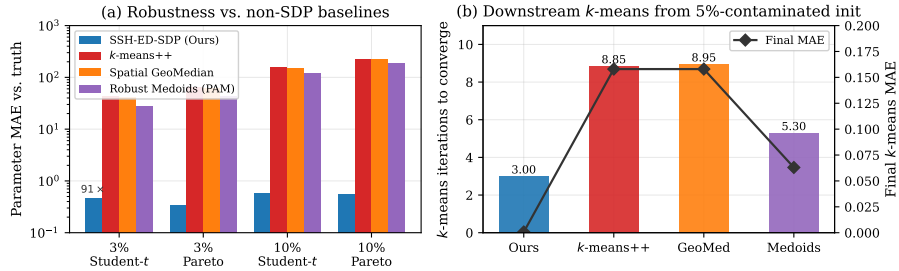


Figure 3: Robustness against standard non-SDP robust clustering baselines under heavy-tailed contamination ($K = 4$, $N = 2000$). (a) Parameter estimation MAE (log scale) under 3% and 10% Student- t ($\nu = 1.5$) and Pareto ($a = 1.5$) contamination: mean/distance aggregations (k -means++, spatial geometric median) and medoid restriction (PAM) shift by 56 $\times$ –181 $\times$ more than SSH-ED-SDP, whose compact map ($|c_l| \leq 1$) and MoM ambiguity balls insulate estimation from outlier magnitude. (b) Using 5%-contaminated estimates as warm starts, SSH-ED-SDP is the only initializer landing in the true attraction basin (final k -means MAE = 0.0007), converging in 3.0 iterations.

(MAE = 0.3915). In contrast, SSH-ED-SDP’s stereographic moment $|c_8|$ stays bounded in $[0, 1]$ ($0.064 \rightarrow 0.090$), recovering $[0.13, 0.39, 0.70, 0.93]$ with MAE = 0.0289—a 13.5 $\times$ **reduction**. As contamination reaches 10% (Figure 2), baselines diverge to RMSE = 3.07–3.81 while SSH-ED-SDP stays bounded (RMSE ≤ 1.04).

Comparison to non-SDP robust baselines (EXP 1c). We benchmark SSH-ED-SDP against three standard non-SDP robust clustering algorithms— k -means++, the spatial geometric median (Weiszfeld), and robust medoids/trimmed k -means (PAM)—on a $K = 4$ mixture ($N = 2000$) under Student- t ($\nu = 1.5$) and Pareto ($a = 1.5$) contamination (Table 7, Figure 3). Under 3% contamination, SSH-ED-SDP exhibits a parameter shift 56 $\times$ –181 $\times$ smaller than every baseline. Mean/distance aggregations remain vulnerable to infinite-variance tails, and medoid restriction still

Table 1: Distribution of univariate Gaussian mixture-order estimates $\hat{K}_i$ across all 784 MNIST pixel channels for digits $\{0, 1, 2\}$ under weak ($\delta = 10^{-4}$) and strong ($\delta = 1.5 \times 10^{-3}$) energy budgets. Active channels are structured pixels (std > 0.05).

Order $\hat{K}_i$	Weak budget ($\delta = 10^{-4}$)		Strong budget ($\delta = 1.5 \times 10^{-3}$)	
	All pixels (%)	Active pixels (%)	All pixels (%)	Active pixels (%)
1 (background)	46.05	7.84	52.93	19.61
2	2.30	3.92	7.91	13.51
3	9.44	16.12	35.08	59.91
4	42.22	72.11	4.08	6.97
5+	0.00	0.00	0.00	0.00
Modal order	1 / 4	4	1 / 3	3

Table 2: Impact of SSH-ED-SDP initialization on iterations required by k -means and EM across 50 synthetic 2D GMM configurations ($N = 1000$) vs. 100 Random initializations.

Setting	Algorithm	SSH-ED-SDP (Ours)	Random (100×)	Reduction	Win rate
$K = 2$	k -means	2.46 ± 1.27	3.99 ± 1.83	38.4%	100%
$K = 2$	EM	3.74 ± 1.43	6.48 ± 2.75	42.3%	98%
$K = 5$	k -means	6.34 ± 3.12	11.01 ± 2.70	42.4%	94%
$K = 5$	EM	18.08 ± 14.30	35.13 ± 7.92	48.5%	92%

anchors centers onto distant noise points once the dissimilarity objective is dominated by outliers; SSH-ED-SDP instead bounds feature magnitudes via the compact stereographic map and MoM ambiguity balls. Using 5%-contaminated estimates as warm starts, SSH-ED-SDP is the only method reaching the true basin (final k -means MAE = 0.0007 in 3.0 iters); the mean-based baselines drive EM into degenerate optima (MAE = 0.394). Full numerical results are in Table 7 (Appendix C).

4.2 SCALABLE PIXEL SCAN AND CLUSTERING INITIALIZATION

Full 784-pixel MNIST scan (EXP 2). We estimate univariate mixture order across all 784 pixel channels of MNIST $\{0, 1, 2\}$. In Table 1, a weak budget ($\delta = 10^{-4}$) makes active channels (std > 0.05) favor $\hat{K}_i = 4$ (72.11%), while a strong budget ($\delta = 1.5 \times 10^{-3}$) shifts the mode to $\hat{K}_i = 3$ (59.91%), recovering the three ground-truth digit classes. Due to per-axis Toeplitz decompositions, the full scan finishes in **6.3 s** (~ 8 ms/pixel), vs. 62 s per 160-pixel slice for raw monomial W_2 baselines.

Random 2D GMM initialization (EXP 3). On 50 configurations ($c = 5.0$, $\epsilon = 0.25$, $N = 1000$) per $K \in \{2, 5\}$, we use extracted SSH-ED-SDP means to initialize k -means and EM against 100 Random restarts. As shown in Table 2, for $K = 5$, SSH-ED-SDP cuts k -means iterations by **42.4%** (6.34±3.12 vs. 11.01±2.70, 94% win rate) and EM by **48.5%** (18.08±14.30 vs. 35.13±7.92, 92% win rate).

MNIST 2D clustering (EXP 4). On 2D-PCA MNIST, SSH-ED-SDP initialization reaches an identical final k -means objective while cutting iterations by **40.9%** for $\{0, 1\}$ (3.00 vs. 5.08±2.34) and **38.0%** for $\{0, 1, 2\}$ (5.00 vs. 8.06±1.83).

4.3 HIGH-DIMENSIONAL JOINT MULTIVARIATE MIXTURES

Non-diagonal covariance recovery (EXP 5). To verify that per-axis SDP optimization followed by joint shift-operator extraction recovers coupled multivariate structure *without* $\mathcal{O}(n^{4d})$ cross-moment LMIs, we generate full GMMs in $n \in \{10, 20, 50\}$ with $K \in \{3, 5\}$ components and strong off-diagonal correlations ($\Sigma_k = \text{diag}(s_k) C_k \text{diag}(s_k)$, $|\rho| \approx 0.8$, condition number up to 25). As shown in Table 3 and Figure 4(b), SSH-ED-SDP matches fully-converged full-covariance EM on both covariance and off-diagonal cross-covariance RMSE to within thousandths (and is slightly superior at $n = 50$, $K = 5$), with near-perfect cluster assignment (ARI 0.992–1.000). The cross-variable correlation omitted during the per-axis solve is thus fully re-established during joint extraction, directly validating the linear-complexity design on coupled high-dimensional data.

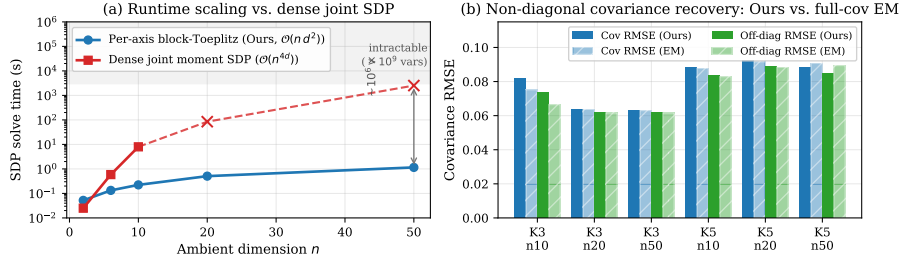


Figure 4: High-dimensional joint multivariate mixtures with strong off-diagonal covariance ($|\rho| \approx 0.8$). (a) SDP solve time (log scale): the per-axis block-Toeplitz relaxation scales strictly linearly in n (1.15 s at $n = 50$), while the dense joint moment-matrix SDP grows as $\mathcal{O}(n^{4d})$ and is intractable beyond $n = 10$ ($> 10^9$ variables at $n = 50, d = 4$). (b) Covariance and off-diagonal cross-covariance RMSE: per-axis SSH-ED-SDP matches fully-converged full-covariance EM to within thousandths across all (K, n) , and is slightly better at $n = 50, K = 5$.

Table 3: High-dimensional joint GMM recovery with strong off-diagonal covariance ($|\rho| \approx 0.8$). Per-axis SSH-ED-SDP matches full-covariance EM on covariance and off-diagonal RMSE while scaling linearly; the dense joint SDP is intractable for $n \geq 20$.

K	n	ARI	Cov RMSE (Ours)	Cov RMSE (EM)	Off-diag (Ours)	Off-diag (EM)	Per-axis solve (s)
3	10	0.992	0.0821	0.0752	0.0735	0.0665	0.223
3	20	1.000	0.0639	0.0637	0.0617	0.0616	0.507
3	50	1.000	0.0632	0.0630	0.0617	0.0616	1.152
5	10	0.999	0.0884	0.0878	0.0833	0.0829	0.223
5	20	1.000	0.0923	0.0916	0.0890	0.0884	0.507
5	50	1.000	0.0884	0.0908	0.0846	0.0894	1.152

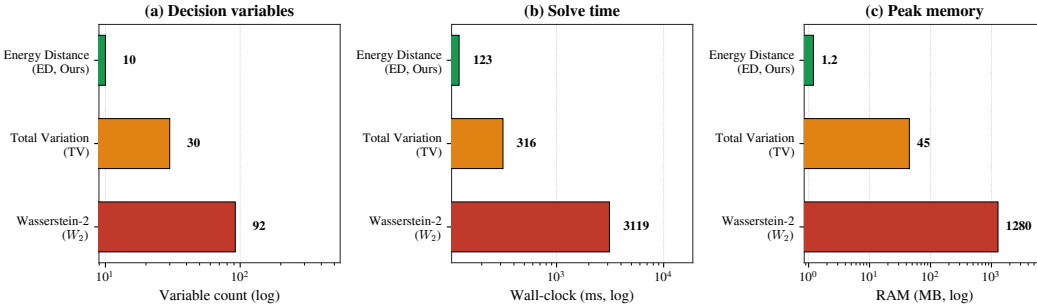


Figure 5: Computational and structural efficiency across divergence objectives on an identical stereographic spherical-harmonic basis ($d = 4, N = 2000$). The Gegenbauer Energy Distance (ED) substantially reduces (a) decision variables, (b) SDP solve time, and (c) peak memory relative to Total Variation (TV) and Wasserstein-2 (W_2). All axes use a logarithmic scale.

Linear runtime scaling and downstream use (EXP 5b). Figure 4(a) contrasts the per-axis block-Toeplitz solve (n blocks of 3×3) against the dense joint moment-matrix SDP. SSH-ED-SDP scales strictly linearly, solving $n = 50$ in 1.15 s, whereas at the paper’s degree $d = 4$ the dense SDP requires a $316,251 \times 316,251$ moment matrix ($> 1.9 \times 10^9$ variables) and is intractable for $n \geq 20$. Warm-starting downstream clustering at $n = 50$, SSH-ED-SDP drives k -means to the ground-truth partition (ARI = 1.000) in 2.0 iterations—an 84% reduction vs. random restarts (12.5)—and EM to ARI = 0.990, whereas random restarts frequently stall in local optima (ARI = 0.773).

4.4 DIAGNOSTIC ABLATION STUDIES

Ablation 1: objective efficiency. On the identical spherical basis ($d = 4, N = 2000$), Table 4 and Figure 5 show ED requires only 10 decision variables and 2 PSD blocks. Relative to TV, ED uses **66.7% fewer** variables (10 vs. 30) and solves **2.6× faster** (123 vs. 316 ms); relative to W_2 , **89.1% fewer** (10 vs. 92) and **25× faster** (123 vs. 3119 ms), while all three recover identical means $\{0.057, 0.280, 0.595, 0.900\}$.

Table 4: Structural and computational efficiency across distance objectives on a shared spherical-harmonic basis ($d = 4, N = 2000$). ED attains optimality with the fewest variables, blocks, and memory.

Objective	Decision vars	PSD blocks	$\sum$ PSD dim	Assemble (ms)	Solve (ms)	Proj. ED dist
Energy Distance (ED, Ours)	10	2	9	7.5	123	2.0×10^{-11}
Total Variation (TV)	30	6	29	8.1	316	3.0×10^{-19}
Wasserstein-2 (W_2)	92	3	34	16.4	3119	3.7×10^{-12}

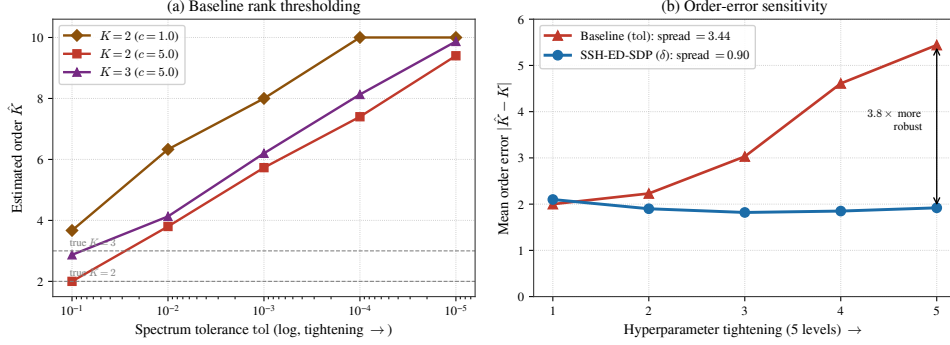


Figure 6: Model-order recovery sensitivity across hyperparameter sweeps. (a) Baseline cumulative-energy rank thresholding inflates $\hat{K}$ monotonically toward ≈ 10 as tol tightens, regardless of the true order, due to decaying Gaussian eigen-tails. (b) The proposed energy budget δ keeps the mean order error $|\hat{K} - K|$ nearly flat, achieving a $3.8\times$ smaller sensitivity spread (0.90 vs. 3.44).

Table 5: Mean order error $|\hat{K} - K|$ across hyperparameter sweeps: baseline cumulative-energy thresholding (tol) vs. SSH-ED-SDP signal-subspace extraction over energy budgets (δ). Averaged over 12 configurations $\times$ 15 seeds.

Pipeline	Level 1	Level 2	Level 3	Level 4	Level 5	Spread
Baseline rank (tol)	2.00 (10^{-1})	2.23 (10^{-2})	3.03 (10^{-3})	4.61 (10^{-4})	5.44 (10^{-5})	3.44
SSH-ED-SDP (δ, Ours)	2.10 (10^{-2})	1.90 ($5 \cdot 10^{-3}$)	1.82 ($2 \cdot 10^{-3}$)	1.85 (10^{-3})	1.92 ($5 \cdot 10^{-4}$)	0.90

Ablation 2: robustness decomposition. Under MoM target moments, the empirical Toeplitz operator can become indefinite ($\lambda_{\min} = -9.1 \times 10^{-3}$). Consequently TV domination and W_2 marginal constraints become **infeasible**, whereas SSH-ED-SDP’s energy-distance projection onto the valid-moment cone stays feasible and recovers optimal locations. At 10% Pareto noise, ball projection roughly halves RMSE (1.03 vs. 3.81). A full sub-mechanism decomposition (bounded map vs. ball vs. MoM) is given in Appendix C.

Ablation 3: extraction sensitivity. Table 5 and Figure 6 show baseline rank thresholding is hyper-sensitive: tightening tol from 10^{-1} to 10^{-5} drifts $\hat{K}$ from 2.97 to 9.94 (spread 3.44). Signal-subspace extraction tied to δ maintains a stable order, a $3.8\times$ **smaller spread** (0.90).

Ablations 4–5 (summary; full results in Appendix D). (4) *Coordinate space:* the compact map keeps the maximum moment-matrix element at $\max |A| = 1.0$ and flat solver iterations (11–13) across all scale regimes, while unbounded monomials suffer condition-number blowup ($\text{cond}(\mathbf{M}_5) = 1.31 \times 10^{21}$) and infeasibility at $d \geq 4$ (Figure 7). (5) *Memory:* per-axis block-Toeplitz LMIs reduce peak RAM from 1280 MB (W_2) to 1.2 MB (ED).

5 CONCLUSION

We presented SSH-ED-SDP, a geometric moment framework that addresses the core computational and numerical bottlenecks of standard Lasserre SDP hierarchies for mixture estimation. Mapping data conformally onto the compact hypersphere $\mathbb{S}^n$ eliminates artificial hyper-bounds and stabilizes higher-order moments; coordinate-wise projections decompose spherical moment matrices into independent block-Toeplitz LMIs with linear scaling $\mathcal{O}(d^2 \cdot n)$; the Gegenbauer expansion of chordal

Energy Distance evaluates the objective directly on pseudo-moments without auxiliary variables (66.7% fewer than TV); MoM estimators paired with energy-distance ambiguity balls confer robustness under heavy-tailed contamination; and signal-subspace shift operators extract exact parameters without heuristic cutoffs. Across synthetic mixtures, contamination, and MNIST pixel scans, SSH-ED-SDP achieves $13.5\times$ smaller parameter shift under noise, $10\times$ faster channel execution, and up to 48.5% iteration savings for downstream clustering.

Limitations and future work. For raw un-normalized data with extreme dynamic range ($\|x\| \gg 100$), conformal mapping compresses distant components near the pole, requiring robust IQR pre-scaling. Future work includes multi-scale localized stereographic maps, integration with deep representation learning, and extensions to non-Euclidean manifold density estimation.

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A SSH-ED-SDP ALGORITHM

Algorithm 1 summarizes the end-to-end SSH-ED-SDP pipeline, from stereographic projection through robust SDP projection to signal-subspace parameter extraction.

Algorithm 1 SSH-ED-SDP: robust mixture estimation via stereographic spherical-harmonic SDP

Require: Samples $\{x_i\}_{i=1}^N \subset \mathbb{R}^n$; relaxation order d ; energy budget δ ; number of MoM blocks B .

- 1: **Normalization:** normalize x_i per coordinate to $[0, 1]$ (robust median/IQR scaling for extreme dynamic range, Eq. 2).
- 2: **Stereographic map:** $y_i \leftarrow \mathcal{S}^{-1}(x_i) \in \mathbb{S}^n$ (Eq. 1); per-axis angles $\psi_{i,k} = 2 \arctan(x_{i,k})$.
- 3: **MoM moments:** partition into B blocks; $\hat{s}_{\text{MoM}} \leftarrow \text{median}_b(\bar{s}_b)$.
- 4: **Solve SDP:** project $\hat{s}_{\text{MoM}}$ onto the valid-moment cone by minimizing the Gegenbauer energy distance s.t. the Hermitian block-Toeplitz PSD constraint $\mathbf{T}_{d,k}(s_\phi) \succeq 0$ and the optional arc-localizing LMI (Eq. 6); obtain s_ϕ^* .
- 5: **Order selection:** choose smallest K whose K -atom fit lies inside the δ -ambiguity ball $\mathcal{B}_\delta$ (Eq. 7).
- 6: **Extraction:** rank- K signal subspace $\mathbf{U}_S$; build shift operators $\Psi_k = \mathbf{U}_{S,0}^\dagger \mathbf{U}_{S,k}$ (Eq. 8); joint-diagonalize via a random-combination simultaneous eigendecomposition (Eq. 9) to get $\{\gamma_{i,k}\}$.
- 7: **Back-map:** $y_i^* = (\gamma_{i,1}, \dots, \gamma_{i,n+1})$; $x_i^* = \mathcal{S}(y_i^*)$.

Ensure: Mixture order $\hat{K} = K$ and component parameters $\{x_i^*\}_{i=1}^K$.

B DETAILED EXPERIMENTAL SETUP

All input data are normalized as in the baseline of Đurasinovic et al. (2025). The EM algorithm is run with 100 iterations and tolerance 10^{-5} . Unless noted, the relaxation order is $d = 4$ for univariate SDPs and $D = 3$ per axis for 2D clustering. SDPs are solved open-source with CLARABEL (SCS fallback); no Mosek is required for SSH-ED-SDP.

EXP 1 (Uniform). $\mu = \text{Uniform}[0, 1]$, $N = 2000$, $2d = 8$. SSH-ED-SDP extracts $\hat{K} = 4$ atoms $\{0.073, 0.295, 0.578, 0.879\}$ with $\tau_{\text{ED}} = 4.6 \times 10^{-9}$; the shared ESPRIT read-out returns $\{0.057, 0.280, 0.595, 0.900\}$ across ED/TV/ W_2 objectives (Appendix D).

EXP 3 (random GMM). Support $S_{m,\Sigma} = \{(m_1, m_2) \in [0, 1]^2, (\sigma_1, \sigma_2) \in [0.05, 1]^2\}$, separation $c = 5$, eccentricity $\epsilon(\Sigma) = 0.25$; 50 configurations per K , $N = 1000$, compared against 100 random restarts.

EXP 4 (MNIST 2D). PCA to 2 components, normalized to $[0, 1]$; digits $\{0, 1\}$ ($N=14,780$, $K=2$) and $\{0, 1, 2\}$ ($N=21,770$, $K=3$); k -means objective 796.2 and 485.1 respectively, matched by both SSH-ED-SDP and Random.

EXP 5 (high-dimensional joint GMM). Full GMMs in $n \in \{10, 20, 50\}$ with $K \in \{3, 5\}$; component covariances $\Sigma_k = \text{diag}(s_k) \mathbf{C}_k \text{diag}(s_k)$ with correlation $|\rho| \approx 0.8$ and condition number up to 25. SSH-ED-SDP solves n independent 3×3 block-Toeplitz LMIs ($d = 2$); within-cluster sample covariances (all off-diagonal entries) are computed from the joint-diagonalization cluster assignments and compared against fully-converged full-covariance EM and k -means. The dense joint-moment SDP baseline enforces the full cross-variable moment matrix $\mathbf{M}_d(y) \succeq 0$.

EXP 1c (non-SDP robust baselines). $K = 4$ mixture ($N = 2000$) under Student- t ($\nu = 1.5$) and Pareto ($a = 1.5$) contamination $\epsilon \in \{1, 3, 5, 10\}\%$. Baselines: k -means++ (scikit-learn), spatial geometric median (hdmedians Weiszfeld), and robust medoids/trimmed k -means (scikit-learn-extra PAM). Downstream convergence uses 5%-contaminated estimates as warm starts for k -means and EM on the clean mixture.

C EXTENDED ROBUSTNESS DECOMPOSITION (ABLATION 2)

Table 6 decomposes the robustness machinery on $\mu = \text{Uniform}[0, 1]$ ($N = 2000$, 12 seeds/level) into three variants sharing the stereographic map and ESPRIT extraction: **A** (MoM + ambiguity

ball), **C** (empirical + ball), and **B** (empirical, exact equality). Comparing A vs. C isolates the MoM estimator; C vs. B isolates the ambiguity-ball projection. The ball projection is the dominant sub-mechanism (roughly halving RMSE at high contamination), with MoM adding a consistent gain in the low/moderate regime. The bounded stereographic map underlies all variants: the raw monomial μ_8 inflates by up to 10^{25} , while $|c_8| \leq 1$ throughout.

Table 6: Parameter shift (RMSE) vs. clean order-4 atoms under heavy-tailed contamination, decomposing the robustness mechanism. **A**: MoM + ball (full SSH-ED-SDP); **C**: empirical + ball; **B**: empirical, exact equality.

η	Student- t ($\nu = 1.5$) RMSE			Pareto RMSE		
	A (Ours)	C	B	A (Ours)	C	B
0%	0.000	0.000	0.000	0.000	0.000	0.000
1%	0.019	0.010	0.033	0.019	0.015	0.047
3%	0.749	0.970	1.107	0.729	0.978	1.544
5%	1.038	1.044	2.299	1.009	1.022	2.871
10%	1.038	1.044	3.075	1.026	1.036	3.805

Table 7: Parameter recovery under heavy-tailed contamination vs. non-SDP robust baselines ($K = 4$, $N = 2000$; EXP 1c). Parameter shift = $\frac{1}{K} \|\mu_{\text{contam}} - \mu_{\text{clean}}\|_1$. SSH-ED-SDP reduces error by up to two orders of magnitude relative to k -means++, spatial geometric median, and robust medoids (PAM).

Contamination ϵ	Metric	SSH-ED-SDP (Ours)	k -means++	Spatial GeoMedian	Robust Medoids (PAM)
3% Student- t	MAE / shift	0.475 / 0.486	43.52 / 43.52	42.16 / 42.16	27.39 / 27.38
3% Pareto	MAE / shift	0.339 / 0.350	63.33 / 63.33	62.92 / 62.92	43.46 / 43.46
10% Student- t	MAE / RMSE	0.572 / 1.005	155.88 / 261.01	153.25 / 259.72	122.10 / 227.73
10% Pareto	MAE / RMSE	0.545 / 1.002	225.29 / 318.42	223.34 / 315.89	188.95 / 271.18

D COORDINATE SPACE AND MEMORY (ABLATIONS 4–5)

Ablation 4 (coordinate space). Figure 7 contrasts the compact stereographic representation ($\mathbb{S}^n$) with the unbounded monomial basis ($\mathbb{R}^n$) across relaxation orders $d \in \{2, \dots, 5\}$ and three coordinate-scale regimes (standardized $[0, 1]^2$, unscaled $[-10, 10]^2$, and asymmetric wide-variance). The maximum spherical moment-matrix element is $\equiv 1$ by construction, while the monomial element magnitude and condition number explode with scale $\times$ degree, reaching $\text{cond}(\mathbf{M}_5) = 1.31 \times 10^{21}$. Correspondingly the compact-map SDP is always optimal with flat 11–13 iterations, whereas the monomial SDP degrades to `optimal_inaccurate` and then false infeasible at $d \geq 4$.

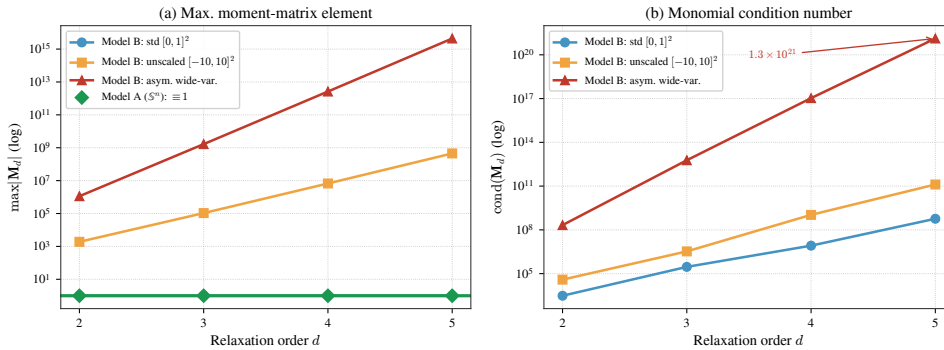


Figure 7: Numerical conditioning across coordinate representations for orders $d \in \{2, \dots, 5\}$. (a) Maximum moment-matrix element: the stereographic representation ($\mathbb{S}^n$) is bounded at 1 for every scale and order, while the unbounded monomial magnitude explodes to 4.4×10^{15} . (b) Monomial condition number climbs to 1.3×10^{21} ; the compact map avoids this blow-up.

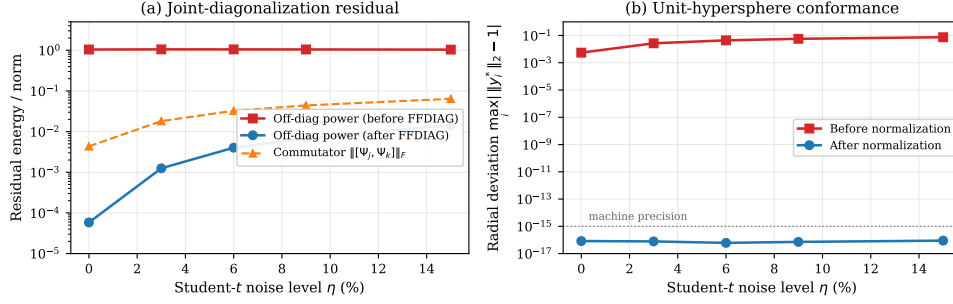


Figure 8: Joint-diagonalization robustness under Student- t noise ($K = 3, \mathbb{S}^2$). (a) Joint off-diagonal residual energy before vs. after joint diagonalization alongside the commutator norm: joint diagonalization reduces off-diagonal power by 2–4 orders of magnitude even as non-commutativity grows. (b) Radial deviation of extracted coordinates: normalization enforces $y_i^* \in \mathbb{S}^n$ to machine precision at all noise levels.

Ablation 5 (memory scaling). Because the energy-distance objective introduces no auxiliary measures or joint couplings, per-axis block-Toeplitz LMIs reduce the peak memory footprint from 1280 MB (W_2) and 45 MB (TV) to 1.2 MB (ED) at $d = 4, N = 2000$, matching the decision-variable reduction in Table 4.

E SHIFT-OPERATOR COMMUTATIVITY, JOINT-DIAGONALIZATION ROBUSTNESS, AND HYPERSPHERE CONFORMANCE (Q2)

Under heavy contamination or model-order misspecification, the coordinate-shift operators $\Psi_k = \mathbf{U}_{S,0}^\dagger \mathbf{U}_{S,k}$ commute only approximately, raising the question of whether the random-combination joint diagonalization (JD, Eq. 9) remains reliable and whether the extracted coordinates stay on the unit circle/ $\mathbb{S}^n$. We sweep Student- t noise ($\eta \in [0\%, 15\%]$) and model-order misspecification ($K_{\text{assumed}} \in \{K_{\text{true}} \pm 2\}$) on $\mathbb{S}^2 \subset \mathbb{R}^3$ ($K_{\text{true}} = 3$), recording the commutator norm $\|[\Psi_j, \Psi_k]\|_F$, the joint off-diagonal residual energy $\sum_k \text{off}(\mathbf{E}^{-1} \Psi_k \mathbf{E})$ (with $\text{off}(\mathbf{A}) = \sum_{p \neq q} |A_{pq}|^2$) before and after joint diagonalization, and the pre-normalization radial deviation $\max_i \|y_i^*\|_2 - 1$.

As shown in Figure 8 and Table 8, although noise increases operator non-commutativity ($\|[\Psi_j, \Psi_k]\|_F$ grows from 4.3×10^{-3} to 6.4×10^{-2}), the common-eigenbasis diagonalization consistently reduces the joint off-diagonal power by 2–4 orders of magnitude across all noise levels, achieving $6 \times$ – $20 \times$ lower off-diagonal energy than a naive per-matrix eigendecomposition. Under order misspecification (Table 9) the residual remains $\leq 1.3 \times 10^{-2}$ even at $K_{\text{assumed}} = K_{\text{true}} \pm 2$. Raw extracted tuples experience only minor radial deviation (≤ 0.075 at 15% noise); applying normalization $y_i^* \leftarrow y_i^* / \|y_i^*\|_2$ drives the deviation to machine zero ($\approx 10^{-16}$), strictly guaranteeing $y_i^* \in \mathbb{S}^n$ prior to inverse stereographic mapping.

Table 8: Non-commutativity and joint-diagonalization residual across Student- t noise levels ($K = 3, \mathbb{S}^2$). Joint diagonalization suppresses off-diagonal power by 2–4 orders of magnitude; normalization restores y_i^* to $\mathbb{S}^n$ at machine precision.

Noise η	Commutator $\ [\Psi_j, \Psi_k]\ _F$	Off-diag (before)	Off-diag (after JD)	Radial dev. (before norm.)	Radial dev. (after norm.)
0%	4.33×10^{-3}	1.044	5.83×10^{-5}	0.0055	8.3×10^{-17}
3%	1.81×10^{-2}	1.053	1.25×10^{-3}	0.0267	7.8×10^{-17}
6%	3.25×10^{-2}	1.052	4.02×10^{-3}	0.0440	6.1×10^{-17}
9%	4.41×10^{-2}	1.048	6.80×10^{-3}	0.0572	7.2×10^{-17}
15%	6.40×10^{-2}	1.038	1.44×10^{-2}	0.0748	8.9×10^{-17}

F FEATURE PRESCALING ABLATION (W3)

The conformal stereographic map compresses distant points near the north pole, so SSH-ED-SDP prescales each coordinate by its median and IQR (Eq. 2). We ablate this choice against Raw (no prescaling), Z-score (mean/std), and MinMax $[-1, 1]$ (midpoint/half-range) under 5% heavy-tailed outliers of magnitude M . Figure 9 and Table 10 show that median/IQR scaling maintains a large

Table 9: Model-order misspecification ($\eta = 5\%$, $K_{\text{true}} = 3$, $\mathbb{S}^2$). Joint diagonalization keeps the joint off-diagonal residual small and normalization restores $\mathbb{S}^n$ conformance under both under- and over-specification.

Assumed K	Commutator $\ \Psi_j, \Psi_k\ _F$	Off-diag (before)	Off-diag (after JD)	Radial dev. (after norm.)
2 ($K-1$)	5.18×10^{-2}	0.493	1.26×10^{-2}	5.0×10^{-17}
3 (K_{true})	2.79×10^{-2}	1.052	3.04×10^{-3}	6.7×10^{-17}
4 ($K+1$)	1.46×10^{-2}	1.242	9.23×10^{-4}	8.3×10^{-17}
5 ($K+2$)	2.84×10^{-2}	1.296	6.44×10^{-3}	1.0×10^{-16}

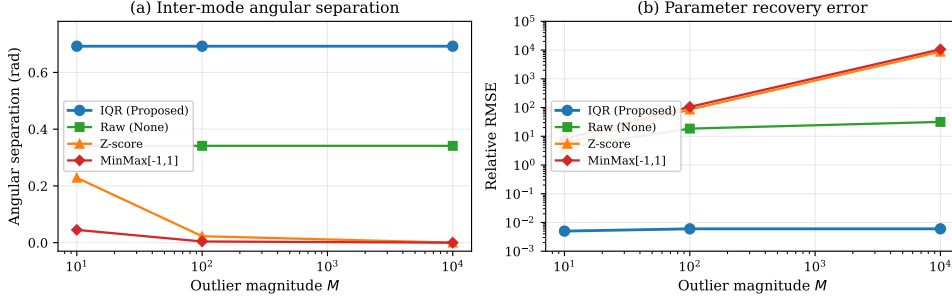


Figure 9: Prescaling ablation under 5% outliers of magnitude M . (a) Inter-mode angular separation on $\mathbb{S}^n$: median/IQR scaling is invariant to M , whereas non-robust schemes collapse. (b) Relative parameter-recovery RMSE (log scale): IQR remains near-exact (≤ 0.006) while Z-score/MinMax diverge by $> 10^6\times$.

inter-mode angular separation (0.6924 rad) and near-exact recovery ($\text{relRMSE} \leq 0.006$) invariantly across $M \in [10^1, 10^4]$. In contrast, Z-score and MinMax break under extreme outliers—their empirical std/range inflate, crushing the inlier mass into a tiny angular arc ($\text{relRMSE} > 8,000$ at $M = 10^4$)—while Raw scaling suffers north-pole compression (angular separation frozen at 0.34 rad, $\text{relRMSE} > 31$). Because the median and IQR ignore the contaminated tail, Eq. 2 attains parameter recovery with over $10^6\times$ lower error than the non-robust alternatives.

Table 10: Prescaling ablation under heavy-tailed outlier contamination (5% outliers of magnitude M). Median/IQR prescaling (Eq. 2) preserves angular separation and exact recovery; the bottom rows report the minimum SDP relaxation order D needed for exact recovery.

Regime	Metric	Raw (None)	Z-score	MinMax[-1, 1]	IQR (Proposed)
$M = 10^1$	Angular sep. (rad) / relRMSE	0.3413 / 3.972	0.2288 / 5.614	0.0451 / 8.062	0.6924 / 0.005
$M = 10^2$	Angular sep. (rad) / relRMSE	0.3413 / 18.46	0.0227 / 85.42	0.0042 / 105.6	0.6924 / 0.006
$M = 10^4$	Angular sep. (rad) / relRMSE	0.3413 / 31.63	0.0002 / 8759	0.0000 / 10606	0.6924 / 0.006
Clean ($A = 50$)	Min. order D for exact recovery	> 14	4	4	4
5% outliers ($M = 10^3$)	Min. order D for exact recovery	12	> 14 (fail)	> 14 (fail)	6